13.11. EXERCISES

13.1. What proof exists that some cosmic rays do not come from the sun?

13.2. (a) What is the primary cosmic radiation hitting earth's atmosphere? (b) Does it penetrate to the earth's surface?

13.3. What background from cosmic radiation is expected for an unshielded 100 ml ion chamber which exhibits an area of 100 cm^2 perpendicular to the direction of the cosmic radiation?

13.4. Which type of mesons are released in high energy particle interactions, and why?

13.5. Cosmic-ray irradiation of the atmosphere yields 0.036 ^{10}Be atoms cm^{-2} s^{-1}. If this ^{10}Be is rapidly carried down into seawater, which is assumed to have a volume of 1.4×10^{18} m^3, what will the equilibrium radioactivity of ^{10}Be in 1 m^3 seawater be? The earth's total surface area is 510×10^6 km^2.

13.6. In Greenland ice the ^{10}Be radioactivity has been measured to be 0.0184 dpm m^{-3}. How old is this ice if it was formed out of water in equilibrium with cosmic-ray ^{10}Be (see previous question)?

13.7. The CO_2 in the atmosphere is in exchange with carbon in living organisms, humus, dissolved organic compounds, and carbonate in the sea water, the latter being the main reservoir (88%) of all exchangeable carbon. The amount of such exchangeable carbon is estimated to be 7.9 g cm^{-2} of the earth. When cosmic-ray produced ^{14}C is mixed into this exchangeable carbon, what will the specific activity become?

13.8. 0.11 cm^3 helium gas at NTP was isolated from 100 g of uranium mineral containing 5 ppm uranium. How old is the mineral?

13.9. A mineral was found to contain 39.1 g K and 87.2×10^{-6} liter Ar at NTP. How old is the mineral?

13.10. A uranium mineral was found to contain the lead isotopes ^{204}Pb, ^{206}Pb, and ^{207}Pb in the ratio 1:1087:388, as determined with a mass spectrometer. Estimate the age of the mineral.

13.11. The heat flow from the earth's crust is 0.060 W m^{-2}. The mean thickness of the crust is 17 km and the earth's radius is 6371 km. The average concentration of uranium, thorium, and potassium in granite is estimated to be 4 ppm (by weight) 18 ppm, and 3.6%, respectively. Assuming that 7% by volume of the crust is made up of granite (feldspar + quartz, density 2.6 g cm^{-3}), what will the heat flow at the earth's surface be from each of these elements? Assume β-heat as 1/3 E_{max}; for α-decay assume $E_\alpha = Q_\alpha$. Each ^{40}K decay by the EC-branch emits one 1.46 MeV γ. Discuss the results.

13.12. A 1 GW_e nuclear power station uses annually about 30 t uranium enriched to 3% ^{235}U. (a) How much natural uranium consumed each year to keep it running? Assume waste stream from isotope separation plant to contain 0.3% ^{235}U.

(b) How much low grade ore (assume 0.06% uranium) must be mined at steady state, if the uranium recovery efficiency in the process is 70%?

13.13. The assumed uranium resources in Japan are 4 kt, in Argentina 12 kt and in France 48 kt U_3O_8. How many 1 GW_e reactor years can these uranium amounts sustain in each country at the uranium consumption rate (a) of the previous exercise?

13.12. LITERATURE

Cosmic radiation

S. Flügge (ed.), Kosmische Strahlung, *Handbuch der Physik,* Band XLV 1/1, Springer-Verlag, Berlin, 1961.
G. Burbidge, The origin of cosmic rays, *Sci. Am.* **215**, Aug. 1966.
A . Webster, The cosmic background radiation, *Sci. Am.* **231**, Aug. 1974.
D. D. Peterman and E. V. Benton, High LET particle radiation inside Skylab command module, *Health Physics* **29** (1975) 125.

Radioelements in nature and dating

E. K. Hyde, *Natural Radioactivity of the Heavy Elements: A Comprehensive Review,* Univ. of California Radiation Laboratory, Report UCRL-10612.
W. F. Libby, *Radiocarbon Dating,* University of Chicago Press, 1956.
C. E. Junge, *Air Chemistry and Radioactivity*, Academic Press, 1963.
H. Craig, S. L. Miller, and G. J. Wasserburg (Eds.), *Isotopic and Cosmic Chemistry*, North-Holland, 1963.
E. I. Hamilton and L. H. Ahrens, *Applied Geochronology*, Academic Press, 1965.
B. Keisch, Dating works of art through their natural radioactivity: improvements and applications, *Science* **160** (1968) 413.
D. York and R. M. Farquhar, *The Earth's Age and Geochronology*, Pergamon Press, 1972.
P. K. Kuroda, Fossil nuclear reactor and plutonium-244 in the early history of the solar system. In IAEA, *The Oklo-phenomenon*, Vienna, 1976.
Ionizing Radiation: *Sources and Biological Effects*, United Nations 1982.
P. S. Zurer, Archeological Chemistry, *Chem. & Eng. News Febr.* **21**, 1983, p. 21.
G. Faure, *Principles of Isotope Geology*, 2nd Ed. J Wiley & Sons 1986.
M. Eisenbud, *Environmental Radioactivity*, 3rd Ed., Academic Press 1987.
M. A. Geyh and H. Schleicher, *Absolute Age Determination*, Springer-Verlag 1990.
G. Brimhall, The Genesis of Ores, *Sci. Am.*, May 1991, p. 48
Fuel Review 1991, Nucl. Eng. Int. Spec. Publ. 1991.
L. Moberg (Ed.), *The Chernobyl Fallout in Sweden*, The Swedish Radiation Protection Institute, Box 60024, S-10401 Stockholm 1991.
T. Nohara, Y. Ochiai, T. Seo and H. Yoshida, PNC, Gify, Japan, Private communication.
Nuclear Geophysics (Int. J. Radiation Appl. and Instrumentation Part E), published by Pergamon Press *World Nuclear Industry Handbook*, Nucl. Eng. Int. Spec. Publ. 2000.